

Introduction

Engineering and technology enable humans to achieve amazing feats. We build skyscrapers that reach toward the clouds, bridges that span great canyons, and tunnels that pierce mountains and travel under the sea. Our computer networks and transportation systems keep people and places connected. We can even explore other planets.

World in motion

Transportation has shrunk our world. Thanks to jet airliners, superhighways, and high-speed rail routes, we can go on long-distance journeys that would have been unthinkable just a few decades ago. This transportation revolution began with the invention of the railroad at the start of the 19th century, and it has continued at speed ever since.

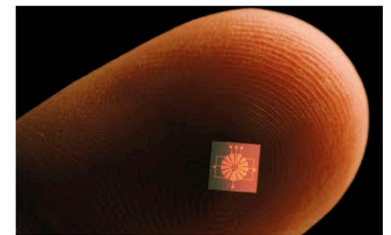
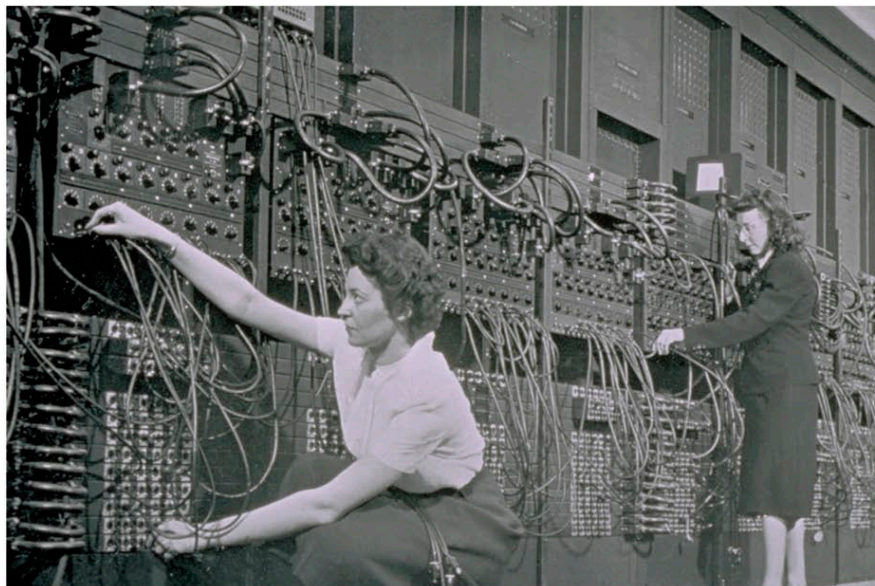
Train collects
electricity from power
cables suspended
above the track.

High-speed electric locomotive

Launched in 1999, the Velaro is now in service in Germany, Spain, France, the UK, China, Russia, Belgium, Turkey, and the Netherlands. It is powered by electricity and can reach speeds of more than 218 mph (350 kph).

Shrinking technology

Few, if any, areas of technology have advanced faster than computing. ENIAC, developed by the US Army in 1946, was the first general-purpose programmable electronic computer. ENIAC contained more than 100,000 components. Since then, electronic components have become smaller and smaller. A modern laptop computer is controlled by a tiny microchip that may be etched with more than a billion components.



Modern marvel

This tiny computer, just 0.04 in (1 mm) square, is implanted into the eye to help people with the disease glaucoma.

Enormous ancestor

ENIAC weighed 33 tons and occupied an entire room. Operators programmed ENIAC by plugging and unplugging cables and adjusting switches.